(CBCP) and Certified Senior Blockchain Professional (CSBCP). CBCP is intended for technical architects who understand blockchain platforms, Bitcoin, business applications of blockchain, cryptocurrency and its usage, development of smart contracts, and design/development of decentralised applications.

CSBCP is advanced professional certification intended to cover the security aspects of the blockchain platform, accounting and smart ledger/ distributed ledger management, regulatory and compliance frameworks across different geo/industry standards, advanced smart contract topics, and financial instrument topics.

BIT also offers various online courses on blockchain, Bitcoin, Ethereum and cryptocurrency.

Blockgeeks certification

Blockgeeks offers various short-term courses coupled with certifications on Ethereum platform topics, blockchain for business professionals (functional and non-technical topics), smart contract security aspects, Hyperledger platform and its tools, and the blockchain regulatory framework programme. These certifications help you to earn a blockchain certified developer title in specific topics like Bitcoin, Hyperledger, Ethereum, and regulatory frameworks.

Blockchain Training Alliance (BTA)

BTA offers professional certifications like:

- BTA Certified Blockchain Business Foundation (CBBF) to cover the basics of blockchain fundamentals as well as business topics in blockchain applications.
- BTA Certified Blockchain Solution Architect (CBSA) to cover solution design and development of blockchain platforms and applications.
- BTA Certified Blockchain Developer Ethereum (CBDE) to develop Ethereum platform based blockchain applications, including programming constructs and practical experience.
- Certified Blockchain Developer Hyperledger Fabric (CBDH) to develop Hyperledger framework and Hyperledger Fabric based applications, including design and development in the platform.
- Certified Blockchain Security Professional (CBSP) to cover advanced security topics like regulatory frameworks, platform security, application security, and data security around various compliance standards.

Exams for these courses are conducted in Pearson VUE centres like for any other professional certification.

Blockchain Council certifications

Blockchain Council offers domain-centric certifications for blockchain professionals like blockchain certified developer, blockchain certified expert, blockchain marketing professional, blockchain HR professional, blockchain certified architect and blockchain finance professional. It also provides specialisation certifications in Corda, Hyperledger, Bitcoin, cryptocurrency and Ethereum (Figure 1).

Technical skills that professionals in blockchain technology need

To take up a blockchain technology based role, it is important to understand various technical aspects and technology frameworks. For instance, one must understand platform security as any failure on this front can cause the entire blockchain based implementation to collapse.

Other blockchain fundamentals that must be understood include basic technical terms such as cryptocurrency, decentralised networks, nodes, mining algorithms, tokens, hash, initial coin offering (ICO), and forks, to name a few. This is important to understand the differences between various blockchain platforms, and choose the right platform and right architecture for implementing the solution for a given problem.

Technical design of a blockchain platform

A blockchain platform includes the following.

Programming constructs like Java, Python and Ruby, which are required to program the consensus and other decentralised applications in the blockchain platform. Some popular platforms like Hyperledger support a number of programming constructs other than Java and Ruby, such as GoLang, C#, JavaScript, to name a few.

Platform security is fundamental to a blockchain, where one has to understand how to implement an unbreakable architecture to make it reliable for consumers by using various standards like CryptoCurrency Security Standards (CCSS).

Data architecture designs the data model, data flow and data structures to be handled in the transaction to make the entire process efficient, cost-effective, and easy to use for customers.

Application security is designing the security considerations required for various components in the application, including interservice communication and messaging, as well as monitoring and measuring the accessibility to the application.

Security algorithms are the work engines of any blockchain platform, and help to strengthen its computational activities.

Customer Identity Access Management (CIAM) platforms can be any third party integration services like PingIdentity, Akamai, and ForgeRock, to name a few. These services help to transform the customer experience, and handle single sign-on and digital identity for the entire platform.

Machine learning algorithms are a set of algorithms that build a predictive learning model to protect the system from potential attacks.